AI Driven Full Stack Development (AI308B)

Moodle MSE2 Submission

Student Details

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- Roll Number: 202401100400200
- Section: C
- Shift: 2nd
- Case Study Name: Student Grievance Management System

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1. GitHub Repository Link

- https://github.com/UpwanSingh/AIFSD_MSE_2

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2. Render Deployment Links

Backend Deployment Link

- https://student-grievance-backend-mfqb.onrender.com

Frontend Deployment Link

- https://student-grievance-frontend-od85.onrender.com

Live Backend Routes

- GET Health: https://student-grievance-backend-mfqb.onrender.com/
- POST Register: https://student-grievance-backend-mfqb.onrender.com/api/register
- POST Login: https://student-grievance-backend-mfqb.onrender.com/api/login
- POST Submit Grievance: https://student-grievance-backend-mfqb.onrender.com/api/grievances
- GET All Grievances: https://student-grievance-backend-mfqb.onrender.com/api/grievances
- GET Grievance by ID: https://student-grievance-backend-mfqb.onrender.com/api/grievances/:id
- PUT Update Grievance: https://student-grievance-backend-mfqb.onrender.com/api/grievances/:id
- DELETE Grievance: https://student-grievance-backend-mfqb.onrender.com/api/grievances/:id
- GET Search by Title: https://student-grievance-backend-mfqb.onrender.com/api/grievances/search?title=xyz

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3. Project Code Used in Development

Backend Code Files

- backend/src/server.js
- backend/src/models/Student.js
- backend/src/models/Grievance.js
- backend/src/routes/authRoutes.js
- backend/src/routes/grievanceRoutes.js
- backend/src/middleware/authMiddleware.js

- backend/package.json
- backend/.env.example

Frontend Code Files

- frontend/src/App.jsx
- frontend/src/main.jsx
- frontend/src/pages/DashboardPage.jsx
- frontend/src/pages/LoginPage.jsx
- frontend/src/pages/RegisterPage.jsx
- frontend/src/components/ProtectedRoute.jsx
- frontend/src/api.js
- frontend/src/styles.css
- frontend/package.json

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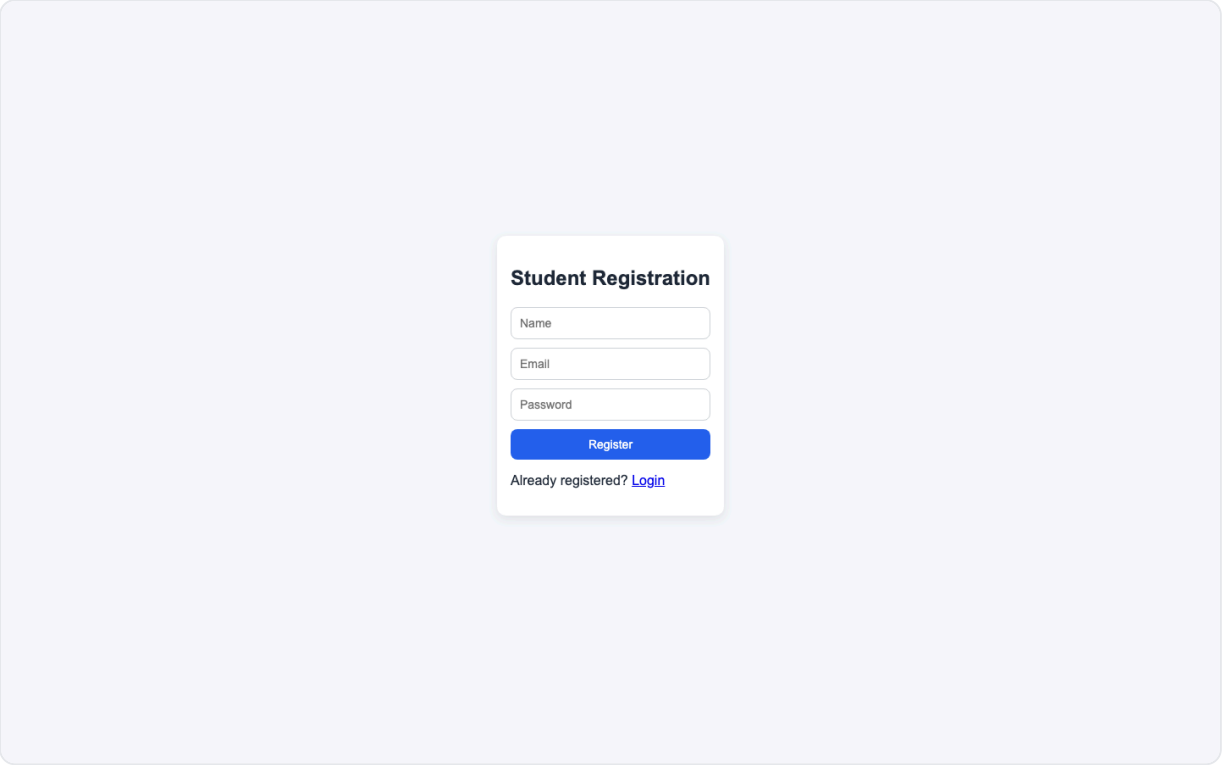
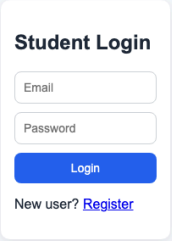
4. Screenshots of Login, Register, Dashboard and All Functional Modules

Auto-captured UI screenshots

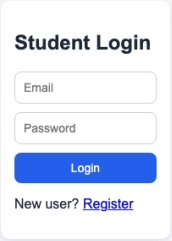
1. Register Page (UI)



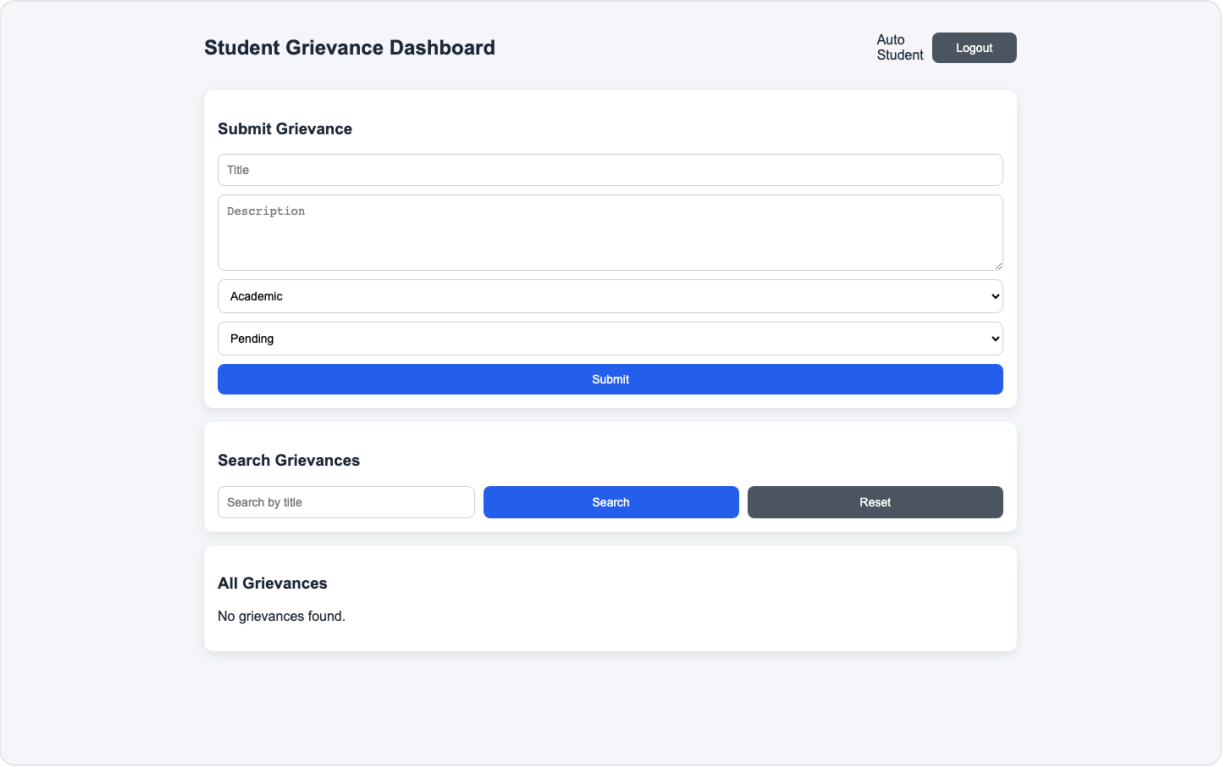
2. Register Success



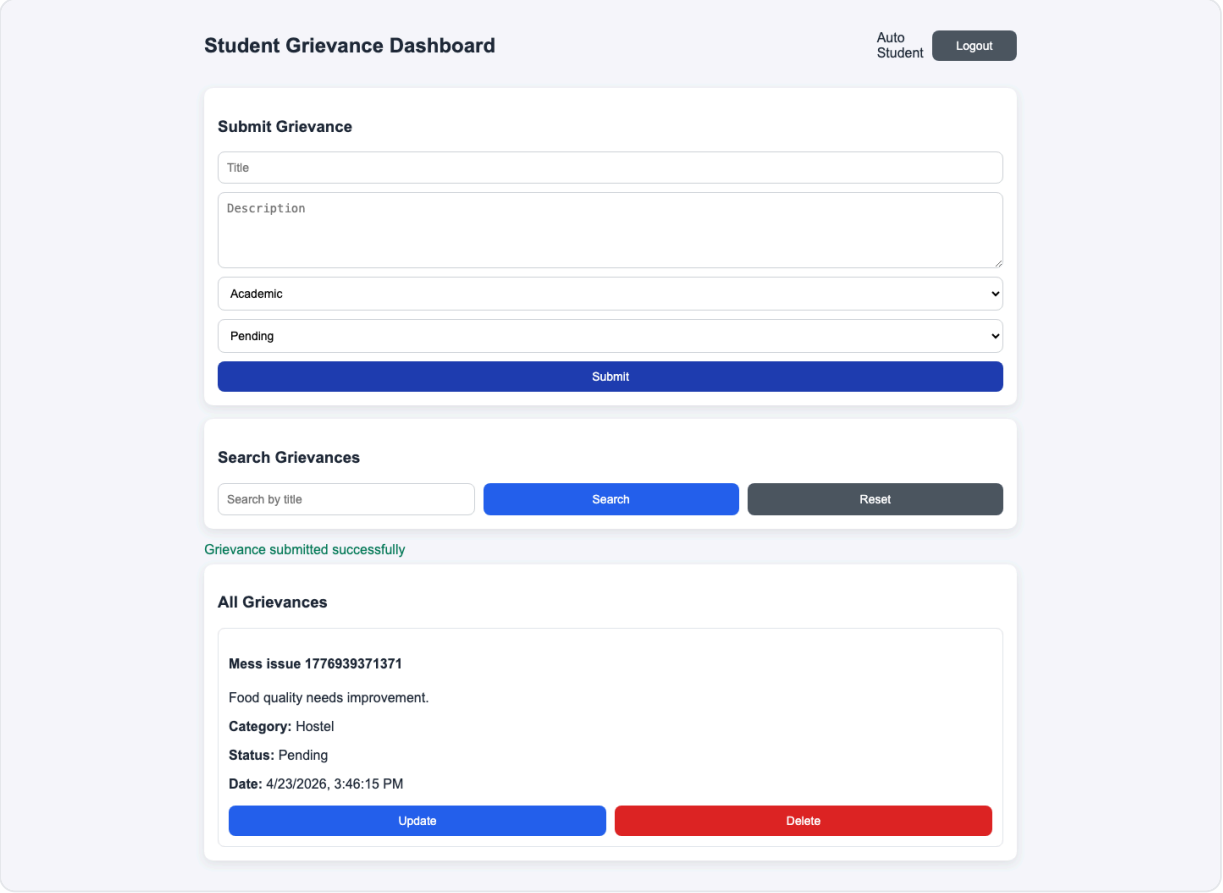
3. Login Page (UI)



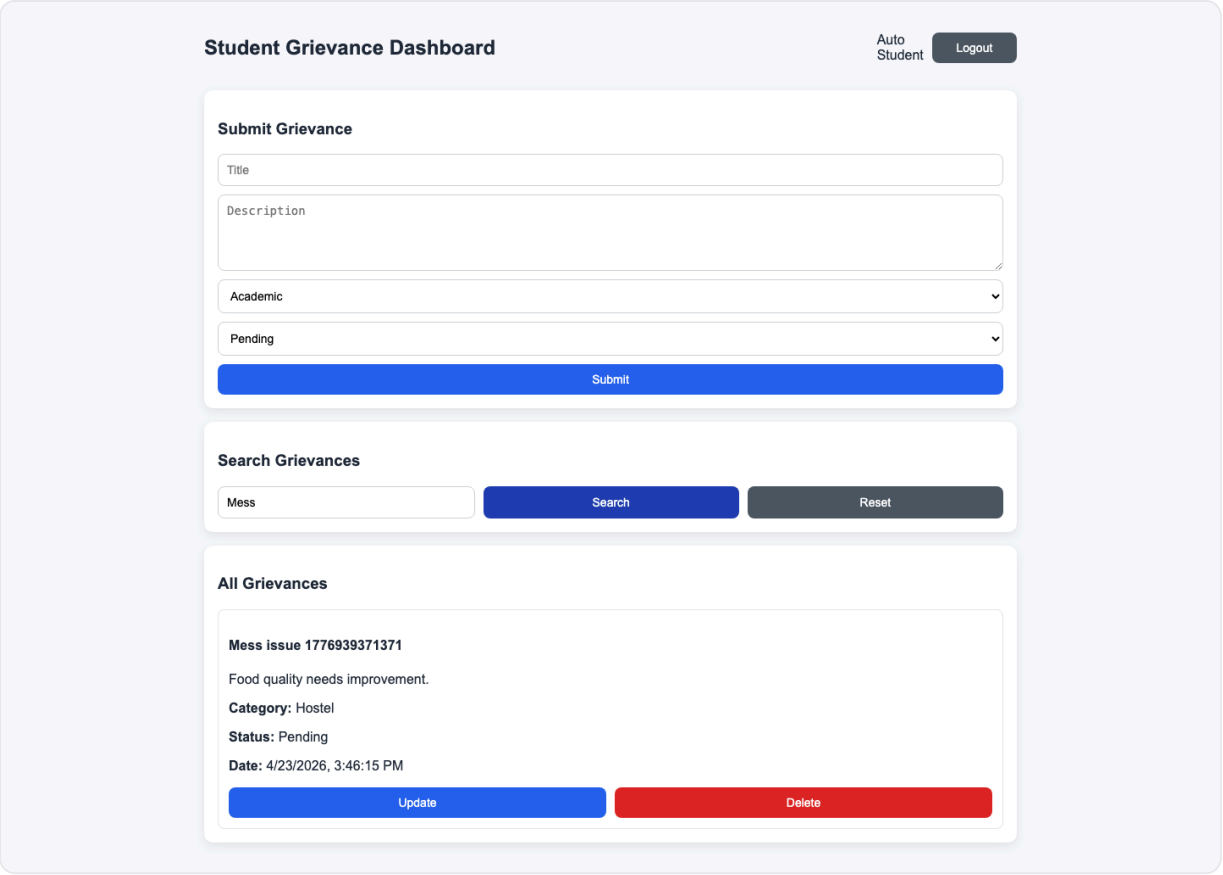
4. Dashboard Page (UI)



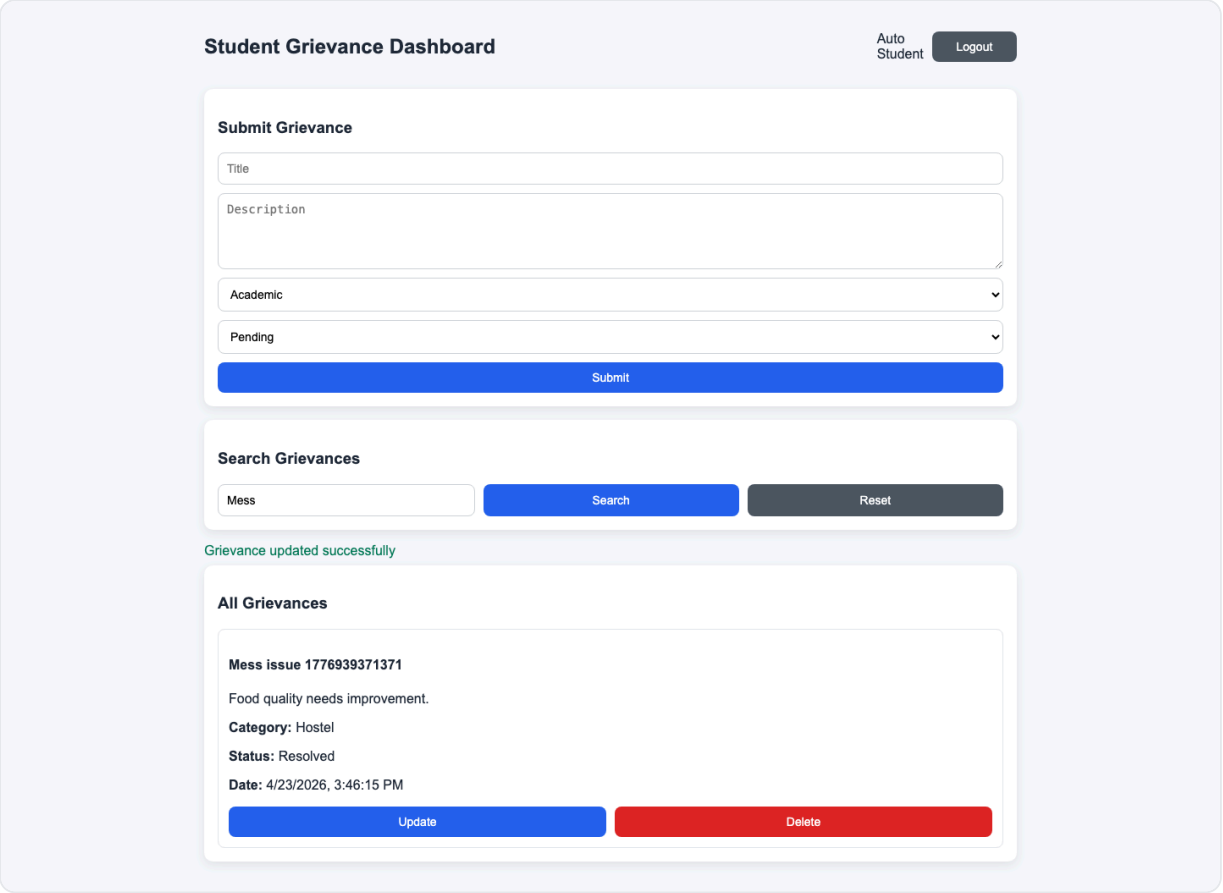
5. Submit Grievance (successful)



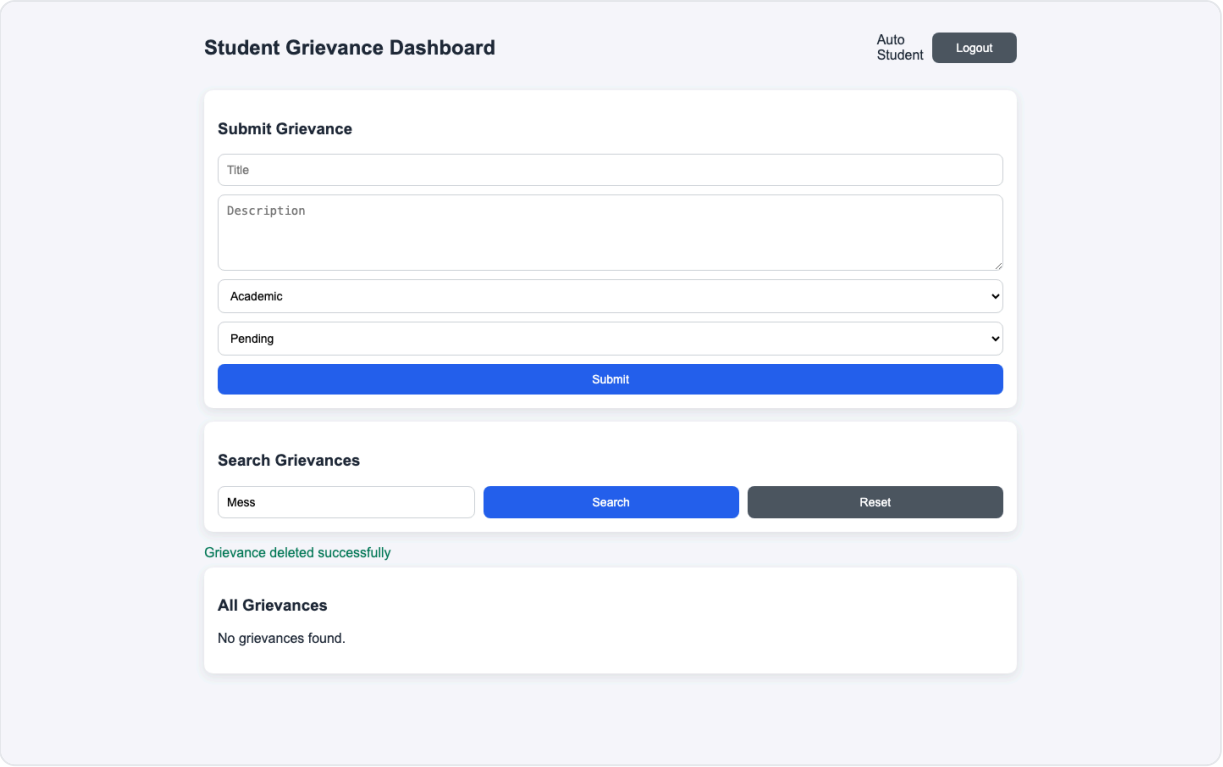
6. Search Grievance



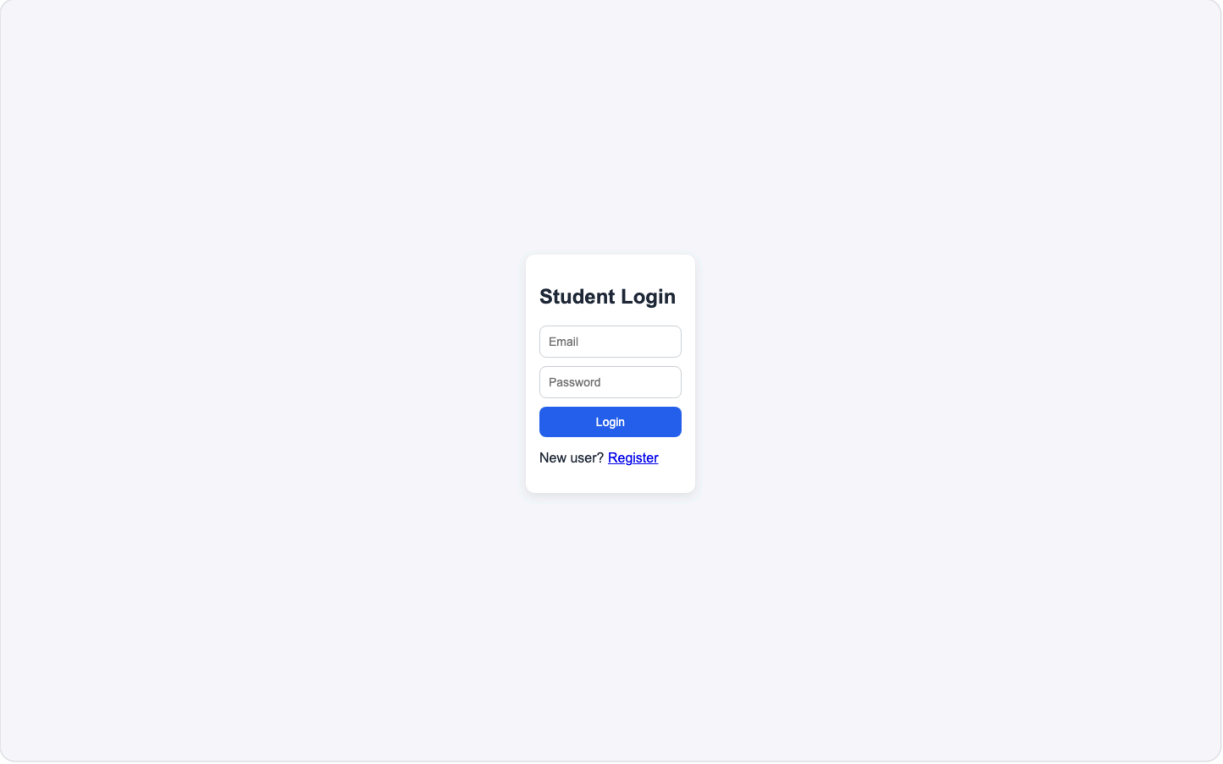
7. Update Grievance



8. Delete Grievance



9. Logout



Add these remaining screenshots manually before final PDF export:

10. API testing screenshots (Postman/Thunder Client)
11. Error handling screenshots:

- Invalid login
- Duplicate email

- Unauthorized access

12. MongoDB data storage screenshot (students + grievances collections)

User-uploaded screenshots found in project folder

13. Uploaded Screenshot 1 Uploaded Screenshot 1

14. Uploaded Screenshot 2 Uploaded Screenshot 2

15. Uploaded Screenshot 3 Uploaded Screenshot 3

API evidence file generated:

- evidence/API_TEST_RESULTS.md

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5. Screenshot of VS Code Project Structure

Add one screenshot from VS Code Explorer showing folder structure including:

- backend
- frontend
- evidence
- render.yaml
- README.md

If one of the uploaded screenshots is the VS Code structure image, keep it under this section in final PDF.

Notes

- Project strictly follows required MERN APIs and functionality from the question paper.
- Password hashing done using bcrypt.
- JWT authentication used for protected grievance routes.
- Deployment configured for both backend and frontend on Render.